Multiple-Choice Questions on Introductory Java Language Features

1. Which of the following pairs of declarations will cause an error message?

```
I.  double x = 14.7;
    int y = x;
II. double x = 14.7;
    int y = (int) x;
III. int x = 14;
    double y = x;
```

(A) I only
(B) II only
(C) III only
(D) I and III only

2. What output will be produced by the following?

```
System.out.print(
    "\\* This is not\n a comment
    *\\");
```

(A) `\* This is not a comment *\`
(B) `This is not`
`a comment *`
(C) `\\* This is not`
`a comment *\\`
(D) `\* This is not`
`a comment *\`

3. Consider the following code segment.

```
if (n != 0 && x / n > 100)
    statement1;
else
    statement2;
```

If `n` is of type `int` and has a value of `0` when the segment is executed, what will happen?

(A) An `ArithmeticException` will be thrown.
(B) ***statement1***, but not ***statement2***, will be executed.
(C) ***statement2***, but not ***statement1***, will be executed.
(D) Neither ***statement1*** nor ***statement2*** will be executed; control will pass to the first statement following the `if` statement.

4. Refer to the following code fragment.

```
double answer = 13 / 5;
System.out.println("13 / 5 = " +
    answer);
```

The output is

```
13 / 5 = 2.0
```

The programmer intends the output to be

```
13 / 5 = 2.6
```

Which of the following replacements for the first line of code will not fix the problem?

(A) `double answer = (double) 13 / 5;`
(B) `double answer = 13 / (double) 5;`
(C) `double answer = 13.0 / 5;`
(D) `double answer = (double) (13 / 5);`

5. Select the value stored in `result` if

```
int result = 13 - 3 * 6 / 4 % 3;
```

(A) −5
(B) 13
(C) −1
(D) 12

6. Suppose that addition and subtraction had higher precedence than multiplication and division. Then the expression

```
2 + 3 * 12 / 7 - 4 + 8
```

would evaluate to which of the following?

(A) 11
(B) 5
(C) 9
(D) −4

7. Which is true of the following boolean expression, given that x is a variable of type double?

```
3.0 == x * (3.0 / x)
```

(A) It will always evaluate to false.
(B) It may evaluate to false for some values of x.
(C) It will evaluate to false only when x is zero.
(D) It will always evaluate to true.

8. Let x be a variable of type double that is positive. A program contains the boolean expression (Math.pow(x,0.5) == Math.sqrt(x)). Even though $x^{1/2}$ is mathematically equivalent to $\sqrt{x}$, the above expression returns the value false in a student's program. Which of the following is the most likely reason?

(A) Math.pow returns an int, while Math.sqrt returns a double.
(B) The computer stores floating-point numbers with 32-bit words.
(C) There is round-off error in calculating the pow and sqrt functions.
(D) There is overflow error in calculating the pow function.

9. What will the output be for the following poorly formatted program segment, if the input value for num is 22?

```
int num = call to a method that reads an
  integer;
if (num > 0)
if (num % 5 == 0)
System.out.println(num);
else System.out.println(num
  + " is negative");
```

(A) 22
(B) 2 is negative
(C) 22 is negative
(D) Nothing will be output.

10. What values are stored in x and y after execution of the following program segment?

```
int x = 30, y = 40;
if (x >= 0)
{
    if (x <= 100)
    {
        y = x * 3;
        if (y < 50)
            x /= 10;
    }
    else
        y = x * 2;
}
else
    y = -x;
```

(A) x = 30 y = 90
(B) x = 30 y = -30
(C) x = 30 y = 60
(D) x = 3 y = -3

11. Which of the following will evaluate to true only if boolean expressions A, B, and C are all false?

(A) !A && !(B && !C)
(B) !(A || B || C)
(C) !(A && B && C)
(D) !A || !(B || !C)

12. Assume that a and b are integers. The boolean expression

```
!(a <= b) && (a * b > 0)
```

will always evaluate to true given that

(A) a = b.
(B) a > b.
(C) a > b and b > 0.
(D) a > b and b < 0.

13. Given that a, b, and c are integers, consider the boolean expression

```
(a < b) || !((c == a * b) &&
(c < a))
```

Which of the following will guarantee that the expression is `true`?

(A) `c < a` is `false`.
(B) `c < a` is `true`.
(C) `c == a * b` is `true`.
(D) `c == a * b` is `true`, and `c < a` is `true`.

14. In the following code segment, you may assume that a, b, and n are all type `int`.

```
if (a != b && n / (a - b) > 90)
{
    /* statement 1 */
}
else
{
    /* statement 2 */
}
/* statement 3 */
```

What will happen if `a == b` is false?

(A) `/* statement 1 */` will be executed.
(B) `/* statement 2 */` will be executed.
(C) Either `/* statement 1 */` or `/* statement 2 */` will be executed.
(D) No statement will be executed.

15. Given that n and `count` are both of type `int`, which statement is true about the following code segments?

I.
```
for (count = 1; count <= n;
count++)
    System.out.println(count);
```

II.
```
count = 1;
while (count <= n)
{
    System.out.println(count);
    count++;
}
```

(A) I and II are exactly equivalent for all input values of n.
(B) I and II are exactly equivalent for all input values $n \geq 1$, but differ when $n \leq 0$.
(C) I and II are exactly equivalent only when $n = 0$.
(D) I and II are not equivalent for any input values of n.

16. The following fragment intends that a user will enter a list of positive integers at the keyboard and terminate the list with a sentinel.

```
int value = 0;
final int SENTINEL = -999;
while (value != SENTINEL)
{
    //code to process value
        ...
    value = ...; //user input
}
```

The fragment is not correct. Which is a true statement?

(A) The sentinel gets processed.
(B) The last nonsentinel value entered in the list fails to get processed.
(C) The code will always process a value that is not on the list.
(D) Entering the `SENTINEL` value as the first value causes a run-time error.

17. Consider this code segment.

```
int x = 10, y = 0;
while (x > 5)
{
    y = 3;
    while (y < x)
    {
        y *= 2;
        if (y % x == 1)
            y += x;
    }
    x -= 3;
}
System.out.println(x + " " + y);
```

What will be output after execution of this code segment?

(A) 1 6
(B) 7 12
(C) 4 12
(D) -3 6

18. Which of the following is a valid comment format in Java?

I. `// This is a valid comment.`
II. `/* This is a valid comment. */`
III. `# This is a valid comment.`

(A) I only
(B) III only
(C) I and II only
(D) I, II, and III

Questions 19–21 refer to the following method, `checkNumber`, which checks the validity of its four-digit integer parameter.

```
/** Returns true if the 4-digit integer n is valid,
 *  false otherwise.
 */
boolean checkNumber(int n)
{
    int d1,d2,d3,checkDigit,nRemaining,rem;
    //strip off digits
    checkDigit = n % 10;
    nRemaining = n / 10;
    d3 = nRemaining % 10;
    nRemaining /= 10;
    d2 = nRemaining % 10;
    nRemaining /= 10;
    d1 = nRemaining % 10;
    //check validity
    rem = (d1 + d2 + d3) % 7;
    return rem == checkDigit;
}
```

A program invokes method `checkNumber` with the statement

```
boolean valid = checkNumber(num);
```

19. Which of the following values of `num` will result in `valid` having a value of `true`?

    (A) `6143`
    (B) `6144`
    (C) `6145`
    (D) `6146`

20. What is the purpose of the local variable `nRemaining`?

    (A) It is not possible to separate `n` into digits without the help of a temporary variable.
    (B) `nRemaining` prevents the parameter `n` from being altered.
    (C) `nRemaining` enhances the readability of the algorithm.
    (D) On exiting the method, the value of `nRemaining` may be reused.

21. Which of the following statements would be an appropriate precondition to add to the comments section above?

    (A) `n > 999 && n < 10000`
    (B) `n > 999 || n < 10000`
    (C) `n >= 999 || n <= 10000`
    (D) `n >= 999 && n <= 10000`

22. For ticket-selling purposes, there are three categories at a certain theater.

Age	Category
65 or above	Senior
From 18 to 64 inclusive	Adult
Below 18	Child

Which of the following code segments will assign the correct string to `category` for a given integer age?

I.
```
if (age >= 65)
    category = "Senior";
if (age >= 18)
    category = "Adult";
else
    category = "Child";
```

II.
```
if (age >= 65)
    category = "Senior";
if (18 <= age <= 64)
    category = "Adult";
else
    category = "Child";
```

III.
```
if (age >= 65)
    category = "Senior";
else if (age >= 18)
    category = "Adult";
else
    category = "Child";
```

(A) I only
(B) III only
(C) II and III only
(D) I, II, and III

23. What output will be produced by this code segment? (Ignore spacing.)

```
for (int i = 5; i >= 1; i--)
{
    for (int j = i; j >= 1; j--)
        System.out.print(2 * j - 1);
    System.out.println();
}
```

(A)
```
9 7 5 3 1
9 7 5 3
9 7 5
9 7
9
```

(B)
```
9 7 5 3 1
7 5 3 1
5 3 1
3 1
1
```

(C)
```
1
1 3
1 3 5
1 3 5 7
1 3 5 7 9
```

(D)
```
1 3 5 7 9
1 3 5 7
1 3 5
1 3
1
```

24. Which of the following program fragments will produce this output? (Ignore spacing.)

```
2 - - - - -
- 4 - - - -
- - 6 - - -
- - - 8 - -
- - - - 10 -
- - - - - 12
```

I.
```
for (int i = 1; i <= 6; i++)
{
    for (int k = 1; k <= 6; k++)
        if (k == i)
            System.out.
                print(2 * k);
        else
            System.out.
                print("-");
    System.out.println();
}
```

II.
```
for (int i = 1; i <= 6; i++)
{
    for (int k = 1; k <= i
    - 1; k++)
        System.out.print("-");
    System.out.print(2 * i);
    for (int k = 1; k <= 6
    - i; k++)
        System.out.print("-");
    System.out.println();
}
```

III.
```
for (int i = 1; i <= 6; i++)
{
    for (int k = 1; k <= i
    - 1; k++)
        System.out.print("-");
    System.out.print(2 * i);
    for (int k = i + 1; k
    <= 6; k++)
        System.out.print("-");
    System.out.println();
}
```

(A) I only
(B) II only
(C) I and II only
(D) I, II, and III

25. Consider this program segment.

```
int newNum = 0, temp;
int num = k;
//k is some predefined integer
    value ≥ 0
while (num > 10)
{
    temp = num % 10;
    num /= 10;
    newNum = newNum * 10 + temp;
}
System.out.print(newNum);
```

Which is a true statement about the segment?

I. If `100 ≤ num ≤ 1000` initially, the final value of `newNum` must be in the range `10 ≤ newNum ≤ 100`.
II. There is no initial value of `num` that will cause an infinite `while` loop.
III. If `num ≤ 10` initially, `newNum` will have a final value of `0`.

(A) I only
(B) III only
(C) II and III only
(D) I, II, and III

26. Consider the method `reverse`.

```
/** Returns n with its digits reversed.
 *   - Example: If n = 234, method reverse
 *   - returns 432.
 *   Precondition: n > 0.
 */
int reverse(int n)
{
    int rem, revNum = 0;

    /* code segment */

    return revNum;
}
```

Which of the following replacements for `/* code segment */` would cause the method to work as intended?

I.
```
for (int i = 0; i <= n; i++)
{
    rem = n % 10;
    revNum = revNum * 10 + rem;
    n /= 10;
}
```

II.
```
while (n != 0)
{
    rem = n % 10;
    revNum = revNum * 10 + rem;
    n /= 10;
}
```

III.
```
for (int i = n; i != 0; i /= 10)
{
    rem = i % 10;
    revNum = revNum * 10 + rem;
}
```

(A) I only
(B) II only
(C) I and II only
(D) II and III only

27. Which best describes the precondition of a method? It is an assertion that

(A) describes precisely the conditions that must be true at the time the method is called.
(B) initializes the parameters of the method.
(C) describes the effect of the method on its postcondition.
(D) states what the initial values of the local variables in the method must be.

28. Refer to the following code segment.

```
/** Compute the mean of integers 1 ... N.
 *  N is an integer >= 1 and has
 *  been initialized.
 */
int k = 1;
double mean, sum = 1.0;
while (k < N)
{
    /* loop body */
}
mean = sum / N;
```

What is the precondition for the while loop?

(A) `k >= N, sum = 1.0`
(B) `k < N, sum = 1.0`
(C) `N >= 1, k = 1, sum = 1.0`
(D) `mean = sum / N`

29. Refer to the following method.

```
/** Precondition: a and b are initialized integers.
 */
public static int mystery(int a, int b)
{
    int total = 0, count = 1;
    while (count <= b)
    {
        total += a;
        count++;
    }
    return total;
}
```

What is the postcondition for method `mystery`?

(A) `total` $= a + b$
(B) `total` $= a^b$
(C) `total` $= a * b$
(D) `total` $= a/b$